

Estimated position sequence

$$\hat{\mathbf{P}} \in \mathbb{R}^{T \times 3}$$

Linear output layer

$$d_{\text{model}} = 128 \rightarrow 3$$

Add & layer normalization

Feed-forward network

$$d_{\text{ff}} = 512, \text{ReLU}$$

Add & layer normalization

Multi-head self-attention

$$N_h = 8, d_k = d_v = 16$$

Transformer encoder layer

$\times 9$; encoder dropout = 0.0534

Sinusoidal positional encoding

$$\text{sequence length } T = 20 \\ \text{dropout} = 0.0534$$

Linear embedding

$$F_{\text{in}} \rightarrow d_{\text{model}} = 128 \\ \text{scaled by } \sqrt{d_{\text{model}}}$$

Input feature sequence

$$\mathbf{X} \in \mathbb{R}^{T \times F_{\text{in}}}; T = 20 \\ F_{\text{in}} = 25 \text{ in simulation}$$